

# MAHIKA MAHAPATRA

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## EDUCATION

### University of Wisconsin–Madison

Sep 2025 – May 2028 (Expected)

B.S. Computer Science | GPA: 3.40 / 4.00 | Dean's List, Spring 2026

**Relevant coursework:** Programming II & III (CS 300/400), Linear Algebra (Math 340), Digital Systems (ECE 252), Statistics (STAT 240), User Interfaces (CS 571), Discrete Mathematics (CS 240), Machine Organization and Programming (CS 354)

## TECHNICAL SKILLS

**Languages:** Java, JavaScript, HTML/CSS, Bash, R, Python, SQL, LC3, C

**Tools & frameworks:** Git/GitHub, React, FastAPI, SQLAlchemy, pandas, JUnit 5, VS Code, Eclipse

## PROJECTS

### RunLoad — Training Load & Injury-Risk Tracker

Jun–Jul 2026

*Python · FastAPI · React · PostgreSQL · pandas | Full-Stack Web App | [github.com/mahikamahapatra101/runload](https://github.com/mahikamahapatra101/runload)*

- Designed and built a full-stack web app (Python/FastAPI, React, SQLAlchemy) that calculates a runner's Acute:Chronic Workload Ratio to flag injury risk.
- Wrote a Python pipeline to clean 1,300+ real runs from a personal Strava export; deployed live on Render and Vercel with a persistent Postgres database.

### ForecastGrid AI — Smart Grid Demand Forecasting

Jun 2026

*R · LSTM Neural Network · Data Engineering · Time-Series Forecasting | FUSE Program, MKE Tech Hub Coalition | "In It To Win It" Award Winner*

- Won the "In It To Win It" Award out of 6 competing teams at FUSE Demo Day hosted by the MKE Tech Hub Coalition.
- Cleaned and merged three real-world datasets (commercial energy, residential consumption, live Milwaukee weather feeds) in R, engineered features, and trained an LSTM neural network to predict energy demand spikes one hour ahead.
- Flagged 165 genuine high-risk demand hours out of 1,643 hours of real Milwaukee-area grid data; built a dashboard prototype delivering automated recommendations to control room operators before spikes occur.

### Stride — Campus Safety Navigation App

Mar 2026

*Java · React · Dijkstra's Algorithm · Safety-Weighted Routing | CheeseHacks 24-Hour Hackathon | [github.com/mahikamahapatra101/stride-cheesehacks](https://github.com/mahikamahapatra101/stride-cheesehacks)*

- Back-end engineer on a 4-person team: designed the graph data model, implemented safety-weighted Dijkstra's to prioritize well-lit paths, and delivered a clean API to the React front-end within 24 hours.
- Pivoted routing logic mid-build after initial edge weighting produced suboptimal paths; recalibrated multipliers on the fly and re-demoed a working product within the same 24 hours.

### Self-Balancing Search Tree

Jan–Mar 2026

*Java · Red-Black Tree · Generics · JUnit 5 | [github.com/mahikamahapatra101/SelfBalancingSearchTree\\_RedBlackTree\\_JUnit](https://github.com/mahikamahapatra101/SelfBalancingSearchTree_RedBlackTree_JUnit)*

- Implemented a generic Red-Black Tree with  $O(\log n)$  guaranteed search and insertion, handling all rotation and color-adjustment cases from scratch with no external libraries.
- Verified all RBT invariants via JUnit 5 across arbitrary object types using Java generics; used as a game leaderboard backend supporting level, playtime, and record-count sorting.

## EXPERIENCE & LEADERSHIP

### Ground Truth Cafe — UW–Madison | Barista

Jan 2026 – Present

- Maintained reliable performance and accuracy in a high-volume cafe while carrying a full CS courseload.

### Science National Honor Society | Co-Founder, Sussex Hamilton High School

2024

- Founded the school's first-ever SNHS chapter: designed the application process, led a 6-month recruitment effort growing to 40+ members, and built all chapter programming with no existing template.

### Varsity Cross Country & Track | Co-Captain · 4-Year Varsity Athlete

2021–2025

- Led team practices and mentored underclassmen as co-captain; qualified for and competed at WIAA State level in Cross Country and Track & Field.

## AWARDS & ACTIVITIES

- FUSE Program: "In It To Win It" Award Winner, MKE Tech Hub Coalition (2026)
- NCWIT Aspirations in Computing: State Winner 2025; State Honorable Mention 2022, 2023, 2024
- National Honor Society: Member, 41+ volunteer hours (2023–2025)
- Girls in STEM, Discovery World: Taught AR and computer vision concepts to middle school students (2023)
- Spanish National Honor Society: Member (2024–2025)